

Lecture Note 13: Comparing Estimators

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In Lecture 5, we introduced mean square error as a means of comparing estimators (i.e., we may prefer an estimator that has a lower mean square error). In this section, we generalize this discussion and analyze the comparison of estimators through the lens of *statistical decision theory*.

1 Basic Concepts

Suppose we observe $X = (X_1, \dots, X_n)'$ i.i.d. realizations from some unknown probability distribution $f(x; \theta)$ for $\theta \in \Theta$. Consider any particular estimator $\hat{\theta} = T(X)$.

To evaluate our estimator, we introduce a *loss function* that associates a non-negative number with any pair (θ, t) , $L(t, \theta)$. For example, for a scalar parameter θ , we could define the loss function as: (i) squared loss, in which case $L(t, \theta) = (t - \theta)^2$; or (ii) absolute loss, in which case $L(t, \theta) = |t - \theta|$. The expected value of the loss function is known as the *risk function*:

$$R(\hat{\theta}, \theta) = \mathbb{E}_\theta[L(\hat{\theta}, \theta)].$$

Example 1. Suppose we observe $X \sim N(\theta, 1)$, and we would like to estimate θ . Consider the squared loss $L(t, \theta) = (t - \theta)^2$ and we consider an estimator of the form $\hat{\theta} = \alpha + \beta X$ for particular values of α, β . We can write the risk for this class of estimators as

$$\begin{aligned} R(\hat{\theta}, \theta) &= \mathbb{E}_\theta[(\alpha + \beta X - \theta)^2] \\ &= V(\hat{\theta}) + \text{Bias}(\hat{\theta})^2 \\ &= \beta^2 V(X) + (\alpha + \beta \mathbb{E}[X] - \theta)^2 \\ &= \beta^2 + (\alpha + (\beta - 1)\theta)^2. \end{aligned}$$

Intuitively, a good estimator should have small risk, and so we would like to select an estimator that solves $\min_{\hat{\theta}} \mathbb{E}_\theta[L(\hat{\theta}, \theta)]$. However, the risk function *depends* on the parameter θ , and so the “best estimator” may depend on the value of θ . But θ is of course unknown to us. Statistical decision theory provides us with alternative criteria to deal with possible tradeoffs in the choice of “best” estimator across values of the unknown parameter θ .

What’s the core problem we face here? The ranking of estimators provided by the risk function is multidimensional – how we rank the performance of estimators varies across possible values of θ . Since θ is unknown to us, we need to somehow aggregate our rankings at each value of θ into a global ranking. There are three main strategies for doing so:

1. Obtain a partial ordering by defining an estimator $\hat{\theta}_1$ to be better than $\hat{\theta}_2$ only if it is better at all possible values of the parameter θ (Admissibility);
2. Obtain a complete ordering by defining an estimator $\hat{\theta}_1$ to be better than $\hat{\theta}_2$ if it is better *on average* across possible values of the parameter (Bayes Risk);

3. Obtain a complete ordering by defining an estimator $\hat{\theta}_1$ to be better than $\hat{\theta}_2$ if it is better under its worst-case scenario (Minimax).

1.1 Admissibility

We say an estimator $\hat{\theta}$ is *inadmissible* or *dominated* if there exists another estimator $\tilde{\theta}$ such that

$$R(\tilde{\theta}, \theta) \leq R(\hat{\theta}, \theta)$$

for all $\theta \in \Theta$ and

$$R(\tilde{\theta}, \theta) < R(\hat{\theta}, \theta)$$

with strict inequality for some θ . Estimators that are not dominated are called *admissible estimators*. Dominance only establishes a partial ordering of estimators – in general, many different estimators are admissible.

Example 2. Suppose we observe $X_i \sim N(\theta, 1)$ i.i.d. for $i = 1, \dots, n$. We would like to estimate θ with squared loss, and we consider two estimators $\hat{\theta}_1 = X_1$ and $\hat{\theta}_{1:n} = \bar{X}$. As in the previous example, we can calculate the risk of these estimators as

$$\begin{aligned} R(\hat{\theta}_1, \theta) &= V(X_1) + \text{Bias}(X_1)^2 = 1, \\ R(\hat{\theta}_{1:n}, \theta) &= V(\bar{X}) + \text{Bias}(\bar{X})^2 = \frac{1}{n}. \end{aligned}$$

So, provided $n > 1$, we see that $R(\hat{\theta}_{1:n}, \theta) < R(\hat{\theta}_1, \theta)$ for all θ , and so $\hat{\theta}_1$ is inadmissible.

Furthermore, admissibility can be a weak criterion. For example, in the same set-up, consider the “two-estimator” $\hat{\theta} = 2$. The two-estimator is admissible under squared loss!

1.2 Integrated Risk and Bayes Estimators

Suppose we introduce a non-negative *weight function* $\omega(\theta)$ on Θ . We define the *integrated risk* of an estimator $\hat{\theta}$ as

$$R(\hat{\theta}, \omega) = \int R(\hat{\theta}, \theta) \omega(\theta) d\theta.$$

We could then select an estimator to minimize integrated risk

$$\min_{\hat{\theta}} R(\hat{\theta}, \omega).$$

Notice that integrated risk and the resulting estimator depend crucially on the choice of the weight function $\omega(\theta)$.

Let us now further assume that the weight function $\omega(\theta)$ is itself a probability density, in which case we refer to $\omega(\theta)$ as a *prior distribution* and $R(\hat{\theta}, \omega)$ is referred to as *Bayes risk*. A *Bayes estimator* minimizes Bayes risk:

$$\hat{\theta}^{\text{Bayes}} = \arg \min_{\hat{\theta}} R(\hat{\theta}, \omega).$$

Given a particular prior distribution $\omega(\theta)$, we can define (i) the joint density of X, θ as $f_{X,\theta}(x, t) = f(x; t)\omega(t)$; (ii) the marginal density of X as $f_X(x) = \int f_{X,\theta}(x, t)dt$, and (iii) the conditional density of θ given $X = x$ as

$$f_{\theta|X}(t | x) = \frac{f_{X,\theta}(x, t)}{f_X(x)}.$$

Therefore, we can now define the posterior risk of an estimator $\hat{\theta}$ given the data $X = x$ as

$$R(\hat{\theta}, \omega | X = x) = \int L(\hat{\theta}, t) f_{\theta|X}(t | x) dt.$$

Theorem 1. *A Bayes estimator $\hat{\theta}^{Bayes}$ can be obtained by minimizing posterior risk $R(\hat{\theta}, \omega | X = x)$ at each possible realization $X = x$.*

Proof. To show this, we notice that Bayes risk can be written as

$$\begin{aligned} R(\hat{\theta}, \omega) &= \int R(\hat{\theta}, \theta) \omega(\theta) d\theta \\ &= \int \mathbb{E}_{\theta}[L(\hat{\theta}, \theta)] \omega(\theta) d\theta \\ &= \int \int L(\hat{\theta}, \theta) f(x; \theta) \omega(\theta) dx d\theta \\ &= \int R(\hat{\theta}, \omega | X = x) f_X(x) dx. \end{aligned}$$

Therefore, defining $\hat{\theta}(x) = \arg \min_t R(t, \omega | X = x)$ yields a Bayes estimator. $\square$

Example 3. Suppose $X_i | \theta \sim N(\theta, 1)$ i.i.d. for $i = 1, \dots, n$ and $\theta \sim N(\tau, \omega^2)$, where τ and ω^2 are known. Let $X = (X_1, \dots, X_n)'$ as before, and so $X | \theta \sim N(\theta \ell, I_n)$, where $\ell = (1, \dots, 1)' \in \mathbb{R}^n$ denotes the vector of ones. The normal-normal densities imply that $(X', \theta)'$ is normally distributed with

$$\begin{pmatrix} X \\ \theta \end{pmatrix} \sim N \left(\begin{pmatrix} \tau \ell \\ \tau \end{pmatrix}, \begin{pmatrix} \omega^2 \ell \ell' + I_n & \omega^2 \ell \\ \omega^2 \ell' & \omega^2 \end{pmatrix} \right).$$

Therefore, we can derive the posterior distribution in closed form. Defining $\lambda = 1/\omega^2$, we know that

$$\theta | X = x \sim N \left(\frac{\lambda}{\lambda + n} \tau + \frac{n}{\lambda + n} \bar{x}, \frac{\omega^2}{1 + n\omega^2} \right).$$

If the loss is squared loss, our previous theorem implies that we can construct the Bayes estimator as simply $\hat{\theta}^{Bayes}(x) = \mathbb{E}[\theta | X = x] = \frac{\lambda}{\lambda + n} \tau + \frac{n}{\lambda + n} \bar{x}$. We can then calculate the risk of the Bayes

estimator at a particular value of θ ; it is simply

$$\begin{aligned} R(\hat{\theta}^{Bayes}, \theta) &= V(\hat{\theta}^{Bayes}) + Bias(\hat{\theta}^{Bayes})^2 \\ &= \frac{1}{n} \left(\frac{n}{\lambda + n} \right)^2 + \left(\frac{\lambda}{\lambda + n} (\tau - \theta) \right)^2. \end{aligned}$$

We can compare this to the risk of the sample average $R(\bar{X}, \theta) = \frac{1}{n}$.

It can be shown in general that Bayes estimators are admissible. We prove this result in the simple case in which Θ has finite support. Suppose that $\Theta = \{\theta_1, \dots, \theta_K\}$, and let $\hat{\theta}^{Bayes}$ denote the Bayes estimator under the prior $\omega(\theta)$ satisfying $\omega_k > 0$ for all $k = 1, \dots, K$ where $\omega_k = P_\omega(\theta = \theta_k)$. In this case, the Bayes risk of any estimator $\hat{\theta}$ is simply the weighted sum

$$R(\hat{\theta}, \omega) = \sum_{k=1}^K \omega_k R(\hat{\theta}, \theta_k).$$

Towards a contradiction, suppose that $\hat{\theta}^{Bayes}$ is inadmissible, meaning that there exists another estimator $\tilde{\theta}$ such that $R(\tilde{\theta}, \theta_k) \leq R(\hat{\theta}^{Bayes}, \theta_k)$ for all $k = 1, \dots, K$ with strict inequality $R(\tilde{\theta}, \theta_{k^*}) < R(\hat{\theta}^{Bayes}, \theta_{k^*})$ for some k^* . Since $\omega_k > 0$ for all k , the strict inequality at θ_{k^*} is preserved upon averaging against the prior, and so

$$R(\tilde{\theta}, \omega) = \sum_{k=1}^K \omega_k R(\tilde{\theta}, \theta_k) < \sum_{k=1}^K \omega_k R(\hat{\theta}^{Bayes}, \theta_k) = R(\hat{\theta}^{Bayes}, \omega).$$

But this contradicts the definition of $\hat{\theta}^{Bayes}$ as a minimizer of Bayes risk. Therefore, $\hat{\theta}^{Bayes}$ must be admissible.

Interestingly, the converse can also be shown (under some additional regularity conditions): for every admissible estimator $\hat{\theta}$, there exists some prior distribution $\omega(\theta)$ such that $\hat{\theta}$ is a Bayes estimator for $\omega(\theta)$. Such a result is known as a *complete class theorem*. We will not dive into the details of this result since you will encounter it in later courses that cover statistical decision theory.

1.3 Minimality

A final strategy for selecting estimators is to “guard against the worst case.” That is, let us define the *minimax risk* as

$$\bar{R}(\hat{\theta}) = \sup_{\theta} R(\hat{\theta}, \theta).$$

The *minimax estimator* is defined to be

$$\hat{\theta}^{minimax} = \arg \min_{\hat{\theta}} \bar{R}(\hat{\theta}).$$

2 The Normal Means Model

The ideas of this lecture — risk, admissibility, Bayes and minimax — come together in a single canonical example. Suppose we observe one draw of a p -dimensional vector

$$Y \sim N(\theta, \sigma^2 I_p),$$

with σ^2 known and $\theta \in \mathbb{R}^p$ unknown. This is the *normal means model*. It looks special but it is everywhere: whenever we have many noisy but approximately normal estimates of many different parameters — an estimated effect for each of p subgroups, a value-added estimate for each of p teachers or judges, a coefficient for each of p regressors in the canonical form of the linear model — standardization puts us here, with Y_j the estimate and θ_j the parameter it estimates. We evaluate estimators by quadratic loss, $L(\theta, a) = \|a - \theta\|^2$.

The natural estimator and its risk. The maximum likelihood (and least squares) estimator is $\hat{\theta}_{ML} = Y$, and its risk is

$$R(\theta, \hat{\theta}_{ML}) = \mathbb{E}_\theta \|Y - \theta\|^2 = p\sigma^2,$$

constant over the whole parameter space.

Shrinkage. Consider shrinking the estimator toward the origin: $\hat{\theta}_c = cY$ for a constant $c \in [0, 1]$. Using $Y = \theta + \sigma V$ with $V \sim N(0, I_p)$,

$$R(\theta, \hat{\theta}_c) = \mathbb{E}_\theta \|c\theta + c\sigma V - \theta\|^2 = (1 - c)^2 \|\theta\|^2 + c^2 p\sigma^2.$$

Minimizing over c gives the oracle weight

$$c^*(\theta) = \frac{\|\theta\|^2}{\|\theta\|^2 + p\sigma^2},$$

with minimized risk $c^*(\theta) p\sigma^2 < p\sigma^2$: *some* shrinkage strictly improves on the MLE at every θ . The catch is that c^* depends on the unknown $\|\theta\|^2$, so $\hat{\theta}_{c^*}$ is not an estimator. This is the multivariate version of the scalar shrinkage argument from Lecture Note 5.

Estimating the shrinkage weight. The unknown in c^* can itself be estimated. Since $\mathbb{E}_\theta \|Y\|^2 = \|\theta\|^2 + p\sigma^2$, the data reveal $\|Y\|^2$, and a natural feasible weight replaces the oracle's denominator with $\|Y\|^2$. The *James-Stein estimator* makes a particular choice of this form:

$$\hat{\theta}_{JS} = \left(1 - \frac{(p-2)\sigma^2}{\|Y\|^2}\right) Y.$$

Theorem 2 (James and Stein, 1961). *Suppose $p \geq 3$. Then, for every $\theta \in \mathbb{R}^p$,*

$$R(\theta, \hat{\theta}_{JS}) = p\sigma^2 - (p-2)^2 \sigma^4 \mathbb{E}_\theta \left[\frac{1}{\|Y\|^2} \right] < p\sigma^2 = R(\theta, \hat{\theta}_{ML}).$$

Sketch of proof. Write $\hat{\theta}_{JS} = Y + g(Y)$ with $g(y) = -(p-2)\sigma^2 y/\|y\|^2$, and expand the risk:

$$R(\theta, \hat{\theta}_{JS}) = \mathbb{E}_\theta \|Y - \theta\|^2 + 2 \mathbb{E}_\theta [(Y - \theta)' g(Y)] + \mathbb{E}_\theta \|g(Y)\|^2.$$

The first term is $p\sigma^2$ and the third is $(p-2)^2\sigma^4\mathbb{E}_\theta[1/\|Y\|^2]$. The middle term is handled by *Stein's identity*: for $Y \sim N(\theta, \sigma^2 I_p)$ and any sufficiently smooth g ,

$$\mathbb{E}_\theta [(Y - \theta)' g(Y)] = \sigma^2 \mathbb{E}_\theta [\nabla \cdot g(Y)],$$

which follows from integration by parts against the normal density. A direct calculation gives $\nabla \cdot g(y) = -(p-2)\sigma^2/\|y\|^2$, so the middle term equals $-2(p-2)^2\sigma^4\mathbb{E}_\theta[1/\|Y\|^2]$, and the three terms combine to the stated risk. The requirement $p \geq 3$ is what makes $\mathbb{E}_\theta[1/\|Y\|^2]$ finite. $\diamond$

The conclusions are striking when set against the earlier sections of this lecture. The MLE is *inadmissible* in the normal means model whenever $p \geq 3$: the James-Stein estimator dominates it everywhere, even though the coordinates of Y are independent and each θ_j appears only in the distribution of its own Y_j . Estimating every mean is a different problem than estimating each mean. At the same time, the MLE is not foolish: its constant risk $p\sigma^2$ is the minimax value, so the MLE is minimax — and since $\hat{\theta}_{JS}$ dominates it, the James-Stein estimator is minimax as well. (The James-Stein estimator is itself inadmissible — replacing the shrinkage factor by zero whenever it is negative, the *positive-part* James-Stein estimator, dominates it — an indication that admissibility in this model is a genuinely delicate property.)

The empirical Bayes interpretation. The form of the James-Stein estimator is best understood through a Bayesian calculation in the normal-normal model of Lecture Note 12. Suppose the means themselves are drawn from a prior,

$$\theta_j \sim N(0, \tau^2) \quad \text{i.i.d.}, \quad Y_j | \theta_j \sim N(\theta_j, \sigma^2).$$

The posterior mean of each θ_j is the familiar precision-weighted shrinkage estimator,

$$\mathbb{E}[\theta_j | Y] = \left(1 - \frac{\sigma^2}{\sigma^2 + \tau^2}\right) Y_j,$$

which is exactly $\hat{\theta}_c$ with $c = \tau^2/(\sigma^2 + \tau^2)$. A Bayesian with known τ^2 would shrink by this factor. Now suppose τ^2 is unknown. Marginally, the Y_j are i.i.d. $N(0, \sigma^2 + \tau^2)$, so $\|Y\|^2/(\sigma^2 + \tau^2) \sim \chi_p^2$, and a direct calculation with the χ^2 density shows

$$\mathbb{E} \left[\frac{(p-2)\sigma^2}{\|Y\|^2} \right] = \frac{\sigma^2}{\sigma^2 + \tau^2}.$$

That is: the James-Stein shrinkage factor is exactly an unbiased estimate of the Bayesian shrinkage factor, with the prior's variance estimated from the observed spread of the p estimates themselves. This is the *empirical Bayes* idea: the ensemble teaches us the prior. When we observe many

parallel problems at once — many subgroups, many teachers, many judges — the cross-sectional distribution of the estimates is informative about the distribution of the underlying parameters, and exploiting that information produces estimators that beat treating each problem in isolation. The James-Stein theorem is the frequentist certificate that this logic works: the resulting estimator dominates the MLE without assuming the prior is correct, or that there is a prior at all.